

INDIAN INSTITUTE OF TECHNOLOGY DHARWAD

Department of Computer Science & Engineering

Assignment 1

Assignment report submitted in fulfillment of the requirements for the degree of
Master of Technology in Computer Science and Engineering

Submitted by

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Abstract

This report presents an experimental study of perceptron-based supervised learning for two major machine learning tasks: multi-class classification and regression. The classification experiments consider three-class two-dimensional datasets in both linearly separable and non-linearly separable cases. A perceptron classifier is implemented from scratch using gradient descent, with logistic (sigmoid) and hyperbolic tangent (tanh) activation functions. Since the perceptron is fundamentally a binary classifier, the three-class classification problem is handled using the One-against-One strategy, which trains three binary classifiers corresponding to Class 1 vs Class 2, Class 1 vs Class 3, and Class 2 vs Class 3. The classification performance is analysed using training-error curves, pairwise and combined decision regions, confusion matrices, accuracy, precision, recall, and F-score.

The regression experiments use a perceptron with linear activation and gradient descent for both univariate and bivariate datasets. The univariate experiment uses one input variable to predict a continuous target, while the bivariate experiment uses two input variables. Both models are evaluated using RMSE and percentage RMSE, along with training-error curves, target-versus-model plots, scatter plots, and regression surfaces. The datasets are divided into 70% training and 30% testing samples.

The experimental results show that the perceptron performs extremely well when the classification classes are linearly separable, achieving 100% test accuracy with both activation functions. For the non-linearly separable dataset, the linear decision boundaries limit performance, with logistic and tanh achieving 86.22% and 88.00% accuracy, respectively. In regression, both models converge rapidly but are unable to accurately represent the underlying non-linear relationships because a single perceptron with linear activation can represent only a linear function. Thus, the experiments demonstrate the influence of data separability, activation function, and model capacity on learning performance.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
RMSE	Root Mean Squared Error
%RMSE	Percentage Root Mean Squared Error
F-Score	F-Measure / F1-Score
3D	Three-Dimensional
X	Input Variable in Univariate Regression
X_1, X_2	First and Second Input Variables in Bivariate Regression
Y	Target / Dependent Variable
$\hat{Y}$	Predicted Target / Model Output

1. Introduction

This assignment presents an experimental study of perceptron-based learning applied to two fundamental machine learning tasks: multi-class classification and function regression. The objective is to implement the learning models from scratch using gradient descent and to study their behaviour on different types of datasets. The experiments are designed to examine how the structure of the data and the choice of activation function affect convergence, decision boundaries, prediction accuracy, and overall model performance.

Techniques Used

For the classification problems, a perceptron classifier is implemented from scratch using gradient descent with two activation functions: logistic (sigmoid) and hyperbolic tangent (tanh). The classification task involves three classes in a two-dimensional feature space. Since a standard perceptron is a binary classifier, the One-against-One strategy is employed to extend it to the three-class problem. This results in three binary classifiers: Class 1 vs Class 2, Class 1 vs Class 3, and Class 2 vs Class 3. The predictions of these pairwise classifiers are combined using voting to obtain the final multi-class decision.

Two classification datasets are considered. The first is a linearly separable dataset, consisting of three well-separated classes with no significant overlap. The second is a non-linearly separable dataset, in which the classes have curved, overlapping distributions..

The classification models are analysed using different types of plots and metrics. These measures allow both the learning behaviour and the final classification performance of logistic and tanh activation functions to be compared, particularly to examine whether a perceptron can successfully form suitable decision boundaries for linearly and non-linearly distributed classes.

For the regression problems, a perceptron regression model with linear activation is implemented using gradient descent. Two regression problems are studied: a univariate regression problem, where one input variable (X) is used to predict a continuous target (Y), and a bivariate regression problem, where two input variables (X_1) and (X_2) are used to predict (Y).

The regression models are evaluated using different types of plots and metrics to examine how closely the learned model represents the underlying data relationship.

Problem Statements

The four experiments conducted in this assignment can therefore be stated as follows:

1. **Linearly Separable Classification:**
   To classify three linearly separable two-dimensional classes using a perceptron trained with gradient descent and to compare logistic and tanh activation functions using decision regions, training-error curves, confusion matrices, accuracy, precision, recall, and F-score.
2. **Non-Linearly Separable Classification:**
   To classify three non-linearly separable two-dimensional classes using a perceptron

trained with gradient descent and to compare logistic and tanh activation functions in terms of their decision regions, training behaviour, confusion matrices, accuracy, precision, recall, and F-score. The experiment also demonstrates the limitations of linear perceptron decision boundaries when the class distributions are non-linear.

3. **Univariate Regression:**
   To implement a perceptron-based regression model for a univariate dataset, where one input variable (X) is used to predict a continuous target (Y). The data is divided into training and testing sets in a 70:30 ratio, and the performance of the model is evaluated using training-error curves, target-versus-model plots, scatter plots, RMSE, and %RMSE.
4. **Bivariate Regression:**
   To implement a perceptron-based regression model for a bivariate dataset, where two input variables (X_1) and (X_2) are used to predict a continuous target (Y). The model is evaluated using average-error curves, three-dimensional regression surfaces, target-versus-model scatter plots, RMSE, and %RMSE.

Overall, the assignment aims to demonstrate how a simple perceptron learns decision boundaries and regression relationships through gradient descent, while highlighting the importance of data separability and model capacity.

2. Experimental Setup

2.1. Dataset

2.1.1. Classification

- Dataset - 1 -
  - It is a strongly linearly separable dataset as there is no overlap which can be observed in Fig. 1a.
  - It consists of 3 classes namely Class 1, Class 2, Class 3 with 500 samples of each class

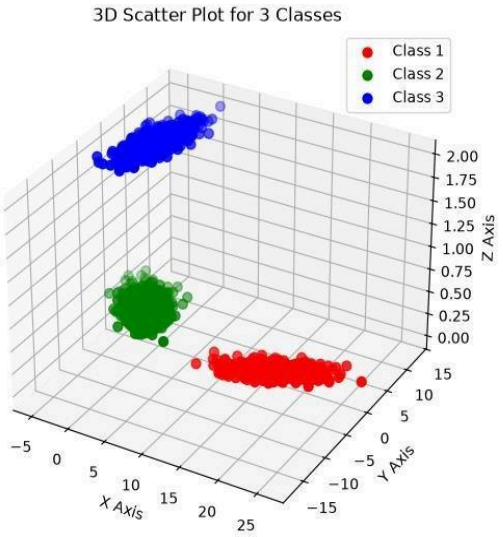


Fig. 1a - 3D scatter plot of Linearly Separable Dataset

- Dataset - 2 -
  - It is a non linearly separable data as there is an overlap between the U shape of the classes which can be observed in Fig. 1b.
  - It consists of 3 classes, namely Class 1, Class 2, Class 3 with each class having 500 samples.

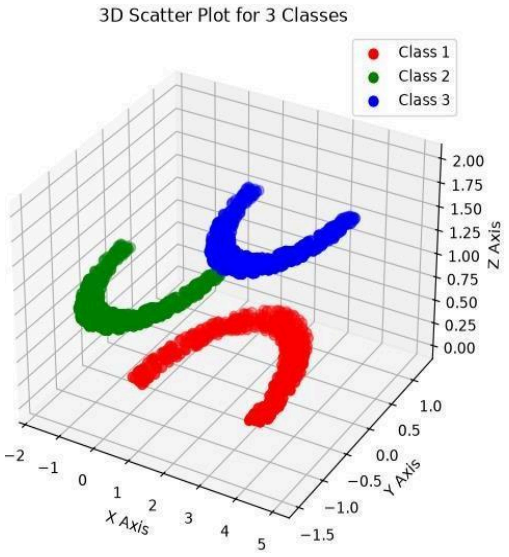


Fig. 1b - 3D Scatter Plot of Non-Linearly Separable Dataset

2.1.2. Regression

- Dataset - 1 -
  - It is a univariate dataset consisting of one independent variable and one dependent variable which can be observed in Fig. 2a.
  - It consists of 1001 samples.

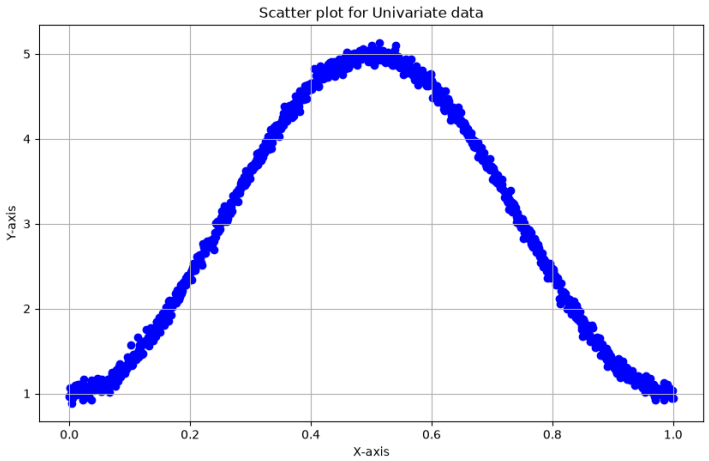


Fig. 2a - Scatter Plot for Univariate Data

- Dataset - 2 -
  - It is a bivariate dataset consisting of two independent variables and one dependent variable which can be observed in Fig. 2b.
  - It consists of 10201 samples.

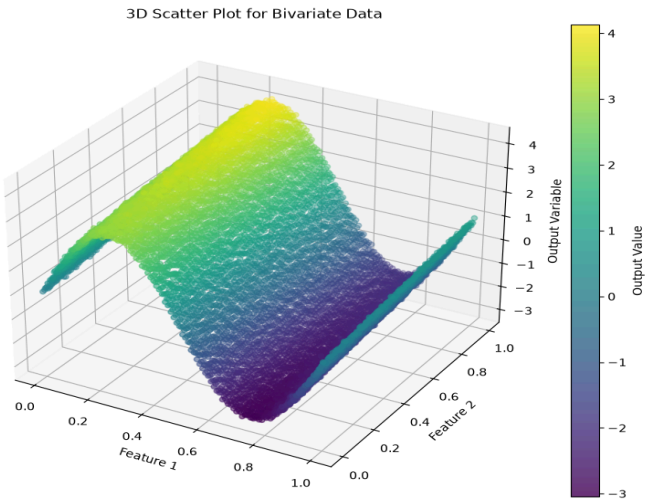


Fig. 2b - 3D scatter plot of Bivariate Data

2.2. Training Set and Test Set

Each of the datasets was partitioned into 30% as test samples and 70% as training samples. The data split uses a fixed random seed of 42 for reproducibility.

Classification Dataset Partition -

The dataset is partitioned in such a way that each class has equal representation in the training set as well as the testing set.

For One-Against-One classification, each binary classifier receives **700 training samples** because it is trained using two classes (350 + 350).
Three classifiers are trained:

- Class 1 vs Class 2
- Class 1 vs Class 3
- Class 2 vs Class 3

The perceptron uses a bias term and gradient-descent weight updates.

Dataset	Train Samples(70%)	Test Samples(30%)	Total Samples
Linearly Separable Dataset	1050 (350 from each class)	450 (150 from each class)	1500
Non-Linearly Separable Dataset	1050 (350 from each class)	450 (150 from each class)	1500

Table 1a - Classification dataset partition

Regression Dataset Partition -

Dataset	Train Samples(70%)	Test Samples(30%)	Total Samples
Univariate Dataset	701	300	1001
Bivariate Dataset	7141	3060	10201

Table 1b - Regression dataset Partition

2.3. Early stopping

Training is checked only after epoch 50 and stops when the change in consecutive average training errors becomes less than 10^{-4} for classification or 10^{-8} for regression .

2.4. Training parameters

- The learning rate is chosen as 0.01 and maximum number of epochs as 1000.
- There was No hyperparameter tuning was performed

3. Experimental Study and Inferences

This section reports training behaviour, classification accuracy, decision regions, and regression fit for each experiment, with an inference drawn from each result.

3.1. Classification

Dataset - 1 (Linearly separable)

- **Average Training Error vs Epochs**
  - **Inference:** The training error decreases rapidly during the initial epochs and then gradually approaches a very small value. This indicates that the gradient-descent learning process is converging. The error becomes sufficiently stable for the early-stopping condition to be satisfied.

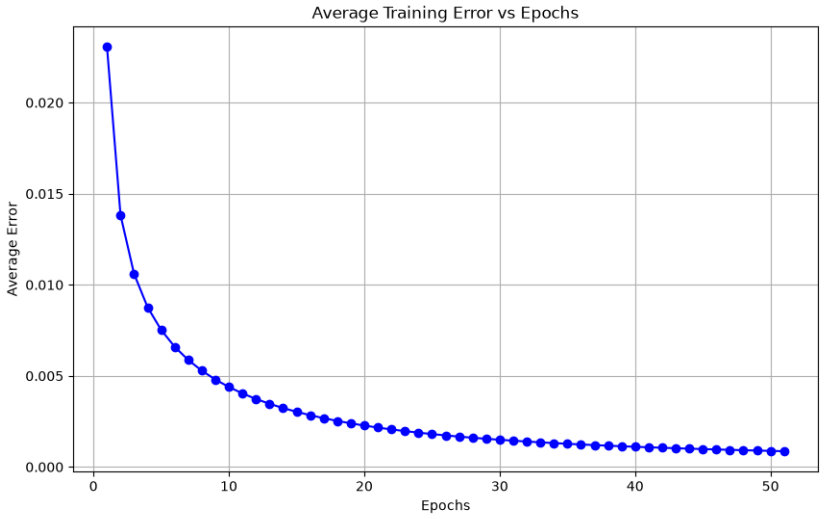


Fig. 3a - Average training error versus epochs for logistic function

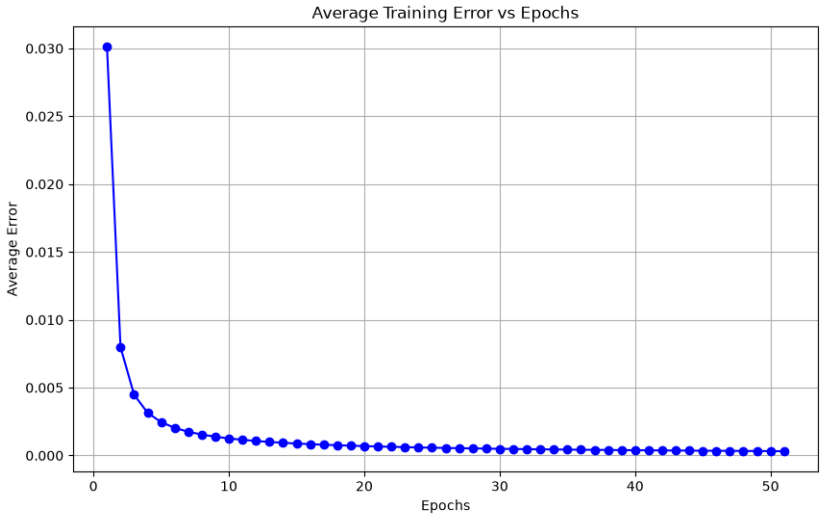


Fig. 3b - Average training error versus epochs for tanh function

  - The tanh activation produced lower final training errors for all three pairwise classifiers in this experiment.

- **Pairwise Decision Regions**
  - **Inference**: The decision-region plots show clear linear boundaries separating every pair of classes. The training samples lie predominantly on the correct side of their respective boundaries. This confirms that each pair of classes can be separated effectively using a linear perceptron decision boundary.

Perceptron Pairwise Result with Tanh function **Perceptron Pairwise Result with logistic function**

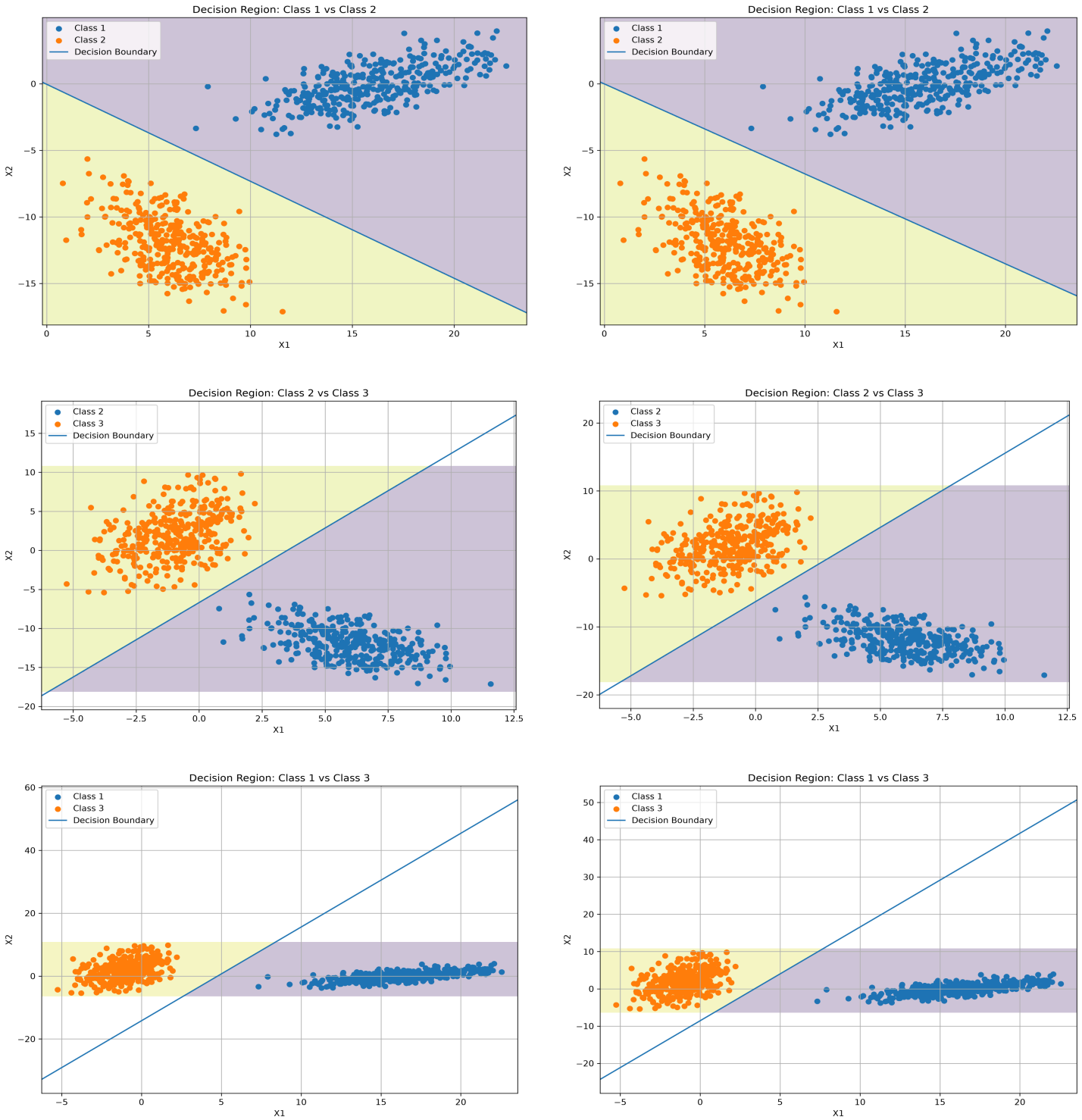
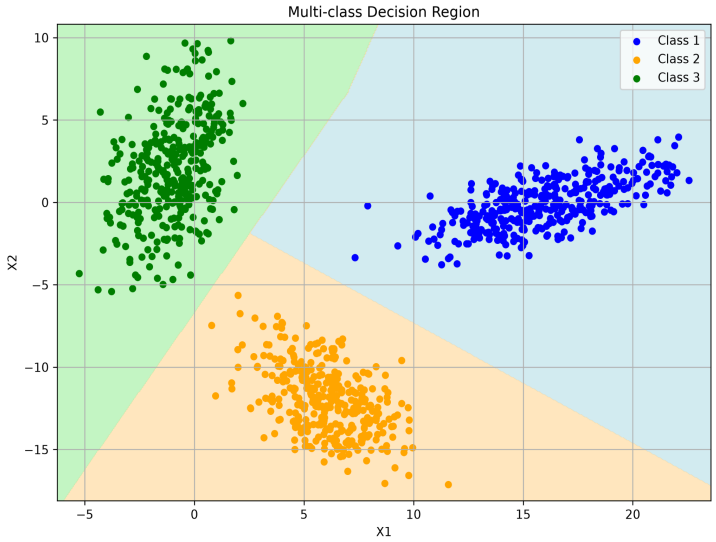


Fig. 4 - Pairwise Decision Region Plot with different activation functions

- **Combined Multi-Class Decision Region**
  - **Inference:** After combining the three One-against-One classifiers, the resulting decision-region plot forms three distinct regions corresponding to the three classes. The training points are located inside their expected class regions, demonstrating that the One-against-One voting mechanism successfully combines the three binary classifiers into a multi-class classifier.

Perceptron Pairwise Result with Tanh function



Perceptron Pairwise Result with logistic function

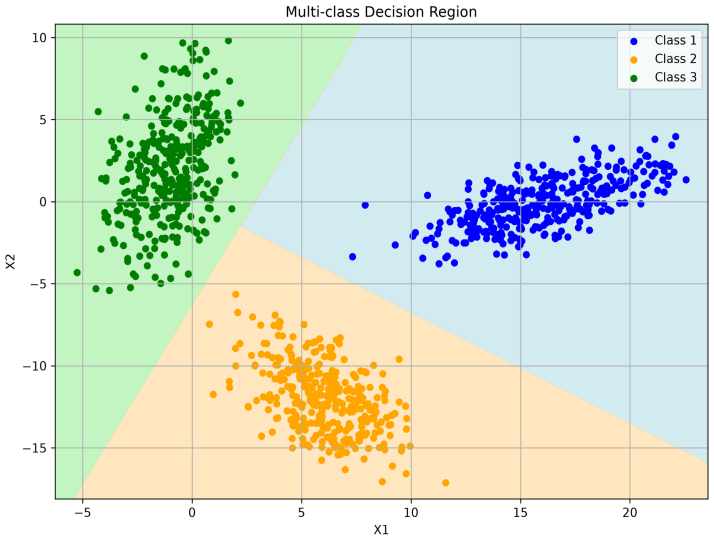


Fig. 5 - Combined Multi-Class Decision Region Plot with different activation functions

- **Inferences on Results Observed**
  - The confusion matrix contains 150 correct predictions for each class and zero misclassifications. Therefore, all 450 test samples were classified correctly.

Metrics	Class 1	Class 2	Class 3
Precision	1.00	1.00	1.00
Recall	1.00	1.00	1.00
F-Score	1.00	1.00	1.00

Table 2 - Metrics vs Classes

  - Accuracy = 1.00 (100%)
  - Macro Precision = 1.0000
  - Macro Recall = 1.0000
  - Macro F-Score = 1.0000

  - The results demonstrate that the perceptron successfully learned the class

boundaries for this linearly separable dataset. The perfect confusion matrix and evaluation metrics are consistent with the clearly separated clusters and decision regions observed in the plots.

- Overall, both logistic and tanh activation functions achieved perfect test classification, while tanh resulted in lower final pairwise training errors in the reported experiment. Since the classes are well separated, neither activation function produced any test misclassification.

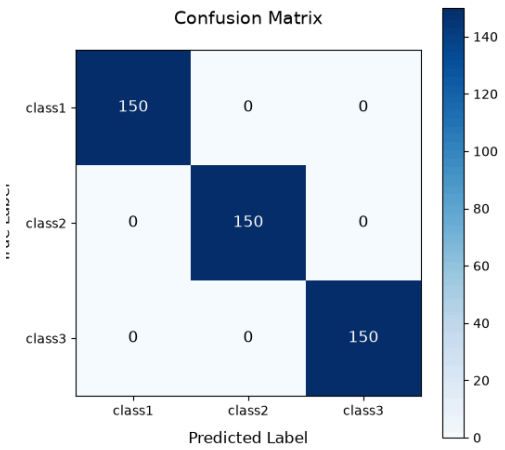
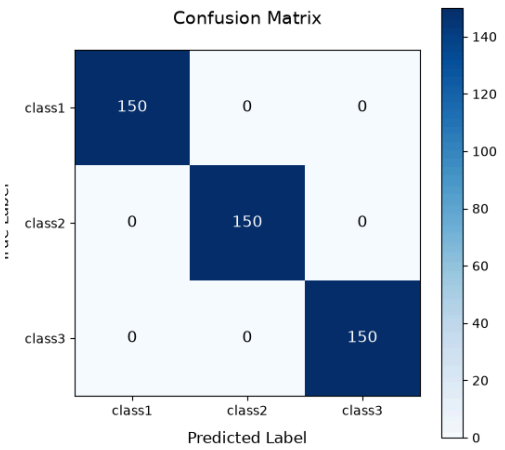
Confusion Matrix for perceptron with tanh function	**Confusion Matrix for perceptron with logistic function**
	

Fig. 6 - Confusion matrices with different activation functions

Dataset - 2 (Non-Linearly separable)

- **Average Training Error vs Epochs**
  - **Inference:** The training-error curves show an overall reduction in error, followed by small variations and a noticeable discontinuity around the early-stopping point. This discontinuity is due to the separate training of the One-against-One binary classifiers and their corresponding error histories rather than a continuous single-model training process. The final errors therefore provide a more meaningful comparison of the individual pairwise classifiers.

Activation	**Classifier**	**Final Error**	**Epochs**
Logistic	Class 1 vs Class 2	0.03241	51
Logistic	Class 1 vs Class 3	0.03139	51
Logistic	Class 2 vs Class 3	0.00728	55
Tanh	Class 1 vs Class 2	0.11788	51

Tanh	Class 1 vs Class 3	0.08261	79
Tanh	Class 2 vs Class 3	0.00293	51

Table 5 - Pairwise Decision Region Plot with different activation functions

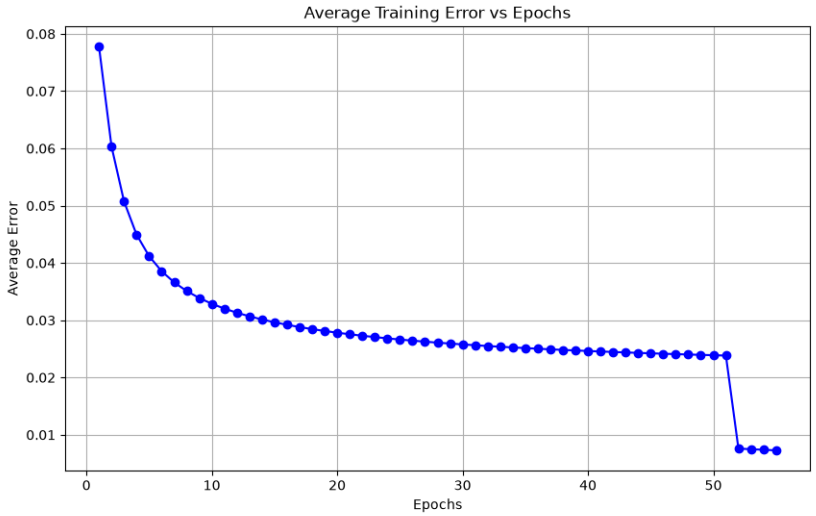


Fig. 7a - Average training error versus epochs for logistic function

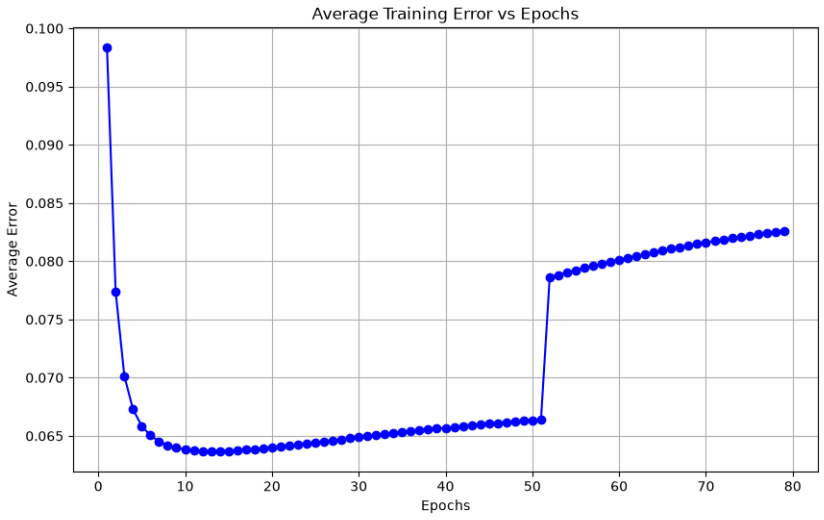


Fig. 7b - Average training error versus epochs for tanh function

- The tanh activation produced lower final training errors for all three pairwise classifiers in this experiment.

- **Pairwise Decision Regions**
  - **Inference**: The pairwise plots between classes show **linear decision boundaries** trying to separate the curved classes. Class 2 vs Class 3 is separated relatively well, while Class 1 vs Class 2 and Class 1 vs Class 3 show greater difficulty and misclassification.

Perceptron Pairwise Result with Tanh function

Perceptron Pairwise Result with logistic function

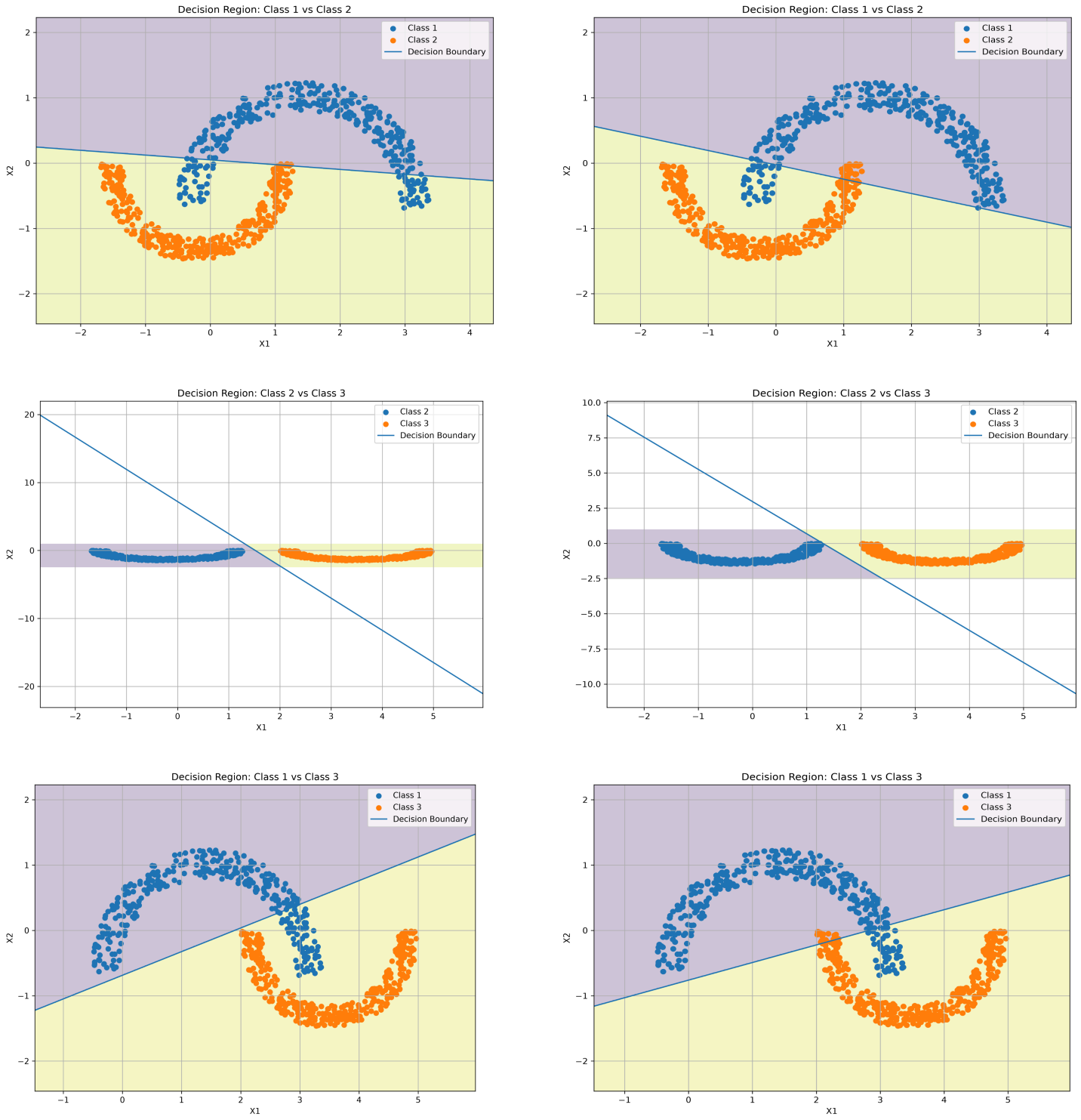
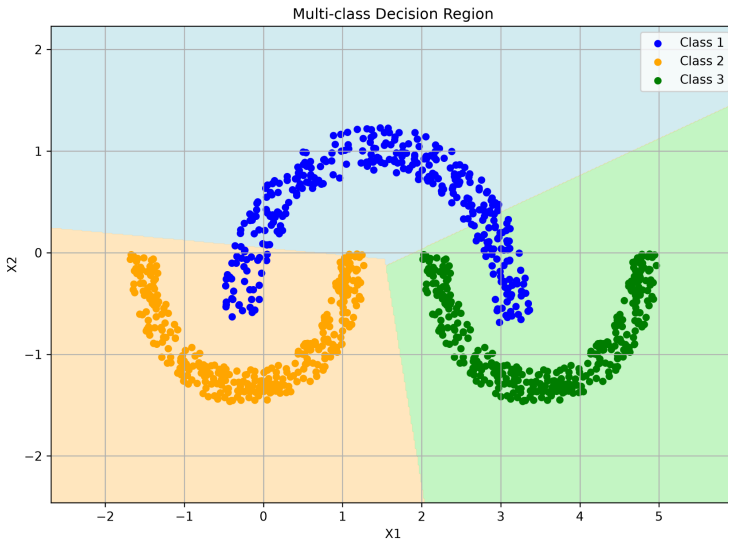


Fig. 8 - Pairwise Decision Region Plot with different activation functions

- **Combined Multi-Class Decision Region**
  - **Inference:** After combining the three One-against-One classifiers, the decision regions form three class regions, but the boundaries do not completely follow the curved data distributions. Consequently, some samples are assigned to incorrect classes.

Perceptron Pairwise Result with Tanh function



Perceptron Pairwise Result with logistic function

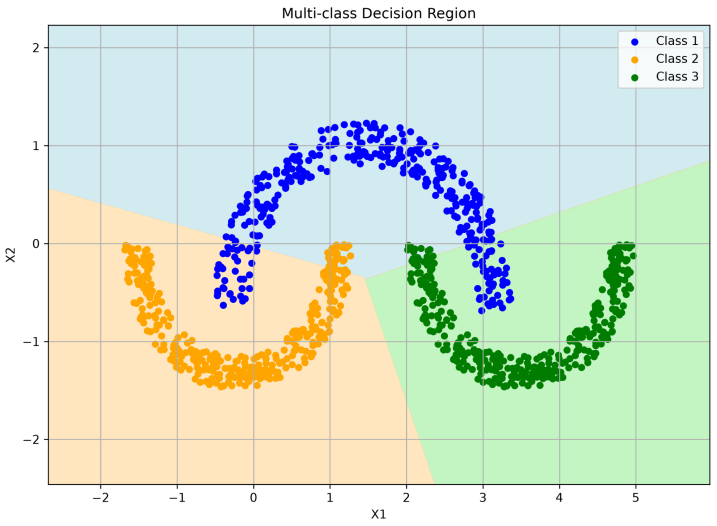


Fig. 9 - Combined Decision Region Plot with different activation functions

- **Inferences on Results Observed**
  - **Logistic Activation**

Metric	Class 1	Class 2	Class 3
Precision	0.85	0.89	0.85
Recall	0.71	0.96	0.91
F-Score	0.78	0.92	0.88

Table 6 - Metrics vs Classes for logistic activation function for non linearly separable data

- **Accuracy = 86.22%**
- **Macro Precision = 0.8613**
- **Macro Recall = 0.8622**
- **Macro F-Score = 0.8589**
- Class 1 has the lowest recall and is the major source of misclassification.

- **Tanh Activation**

Metric	Class 1	Class 2	Class 3
Precision	1.00	0.89	0.81
Recall	0.64	1.00	1.00
F-Score	0.78	0.94	0.90

Table 7 - Metrics vs Classes for tanh activation function for non linearly separable data

  - **Accuracy = 88.00%**
  - **Macro Precision = 0.8995**
  - **Macro Recall = 0.8800**
  - **Macro F-Score = 0.8721**
  - The tanh model correctly classified all test samples belonging to Classes 2 and 3, while Class 1 produced most errors.

- **Overall Comparison**

Metric	Logistic	Tanh
Accuracy	86.22%	88.00%
Macro Precision	0.8613	0.8995
Macro Recall	0.8622	0.8800
Macro F-Score	0.8589	0.8721

Table 8 - Comparison table between logistic and tanh for non linearly separable data

  - **Inference:** Tanh performed slightly better overall, achieving **88% accuracy** compared with **86.22% for logistics**. However, both models show limitations because the perceptron produces linear decision boundaries for a non-linearly separable dataset.
  - Overall, the experiment demonstrates that changing the activation function affects training and classification performance, but a **single-layer perceptron cannot effectively learn the required non-linear decision boundaries**.

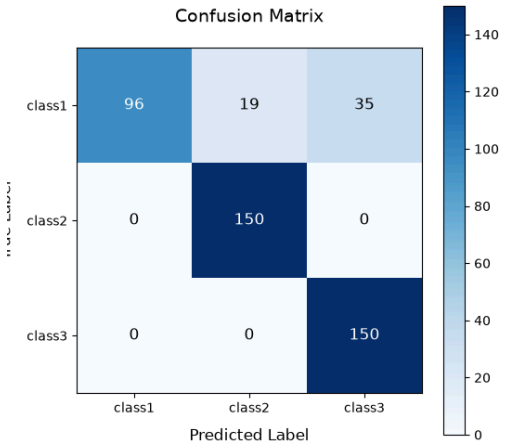
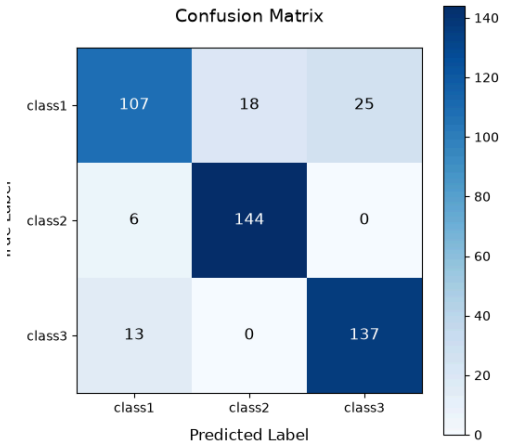
Confusion Matrix for perceptron with tanh function	Confusion Matrix for perceptron with logistic function
	

Fig. 10 - Confusion Matrices with different activation functions

3.2. Regression

Dataset - 1 (Univariate Data)

- **Final Model Weights**
  - The final weights obtained after training were:
    Final weights:
    `[2.752302, 0.01827663]`
  - Thus, the learned regression function is approximately:
    $\hat{Y} = 2.752302 + 0.01827663*X$
  - The very small coefficient of (X) indicates that the trained linear model produces only a small change in output as (X) varies.

- **Training and Testing Results**

Dataset	RMSE	% RMSE
Training	1.43098	47.8355%
Testing	1.43866	47.6899%

Table 9 - Training and Testing results of univariate dataset

  - The training and testing RMSE values are very close. This indicates that the model behaves similarly on both datasets and there is no major difference between training and testing performance.

  - However, the relatively high RMSE and %RMSE indicate that the learned linear perceptron does not model the variation in the target values accurately

since a single perceptron is unable to estimate the function.

- **Average Training Error vs Epochs**
  - **Inference:**
    - The average error decreases sharply during the initial epochs.
    - After the first few epochs, the error becomes almost constant around 1.00. Further training showed a very little improvement.This indicates that the model converges quickly to a stable solution.

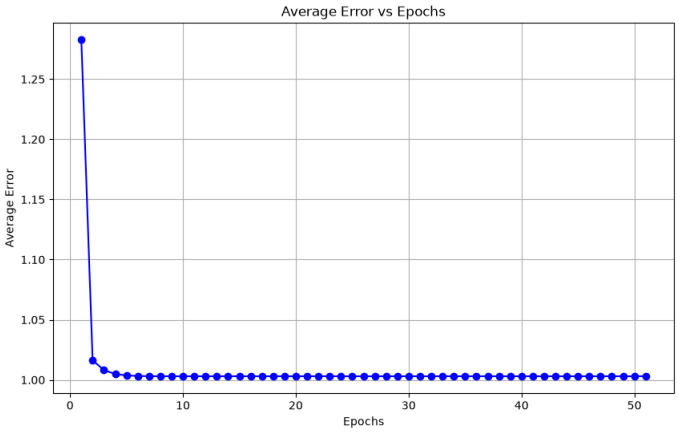


Fig. 11 - Average training error versus epochs

- **Training Target vs Model Output**
  - **Inference**:
    - The actual target values form a clear curved pattern, increasing towards the middle and decreasing afterwards.
    - The model output is a horizontal straight line around 2.75 - 2.77.
    - Therefore, the model fails to capture the nonlinear/curved relationship between (X) and (Y) leading to high RMSE, average error.
    - A linear perceptron cannot reproduce the rise and fall of the target function present in the training data.

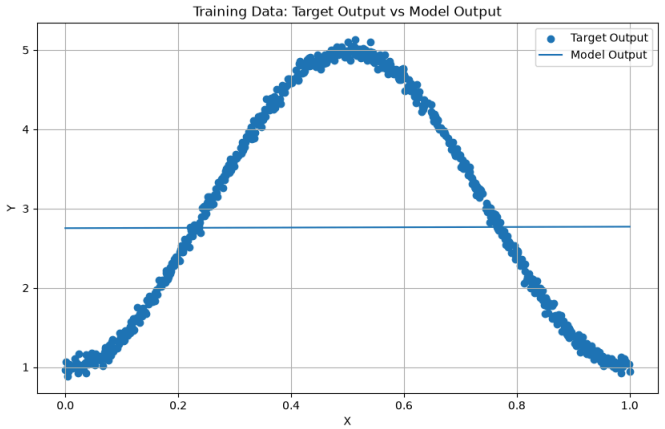


Fig. 12 - Training Target vs Model Output Plot

- **Testing Target vs Model Output**
  - **Inference:**
    - The testing data follows the same curved trend observed in the training data.
    - The model output remains almost constant near 2.75 - 2.77.
    - Therefore, it underestimates the target near the peak.
    - The similar behaviour in training and testing indicates that the problem is primarily model limitation rather than overfitting.

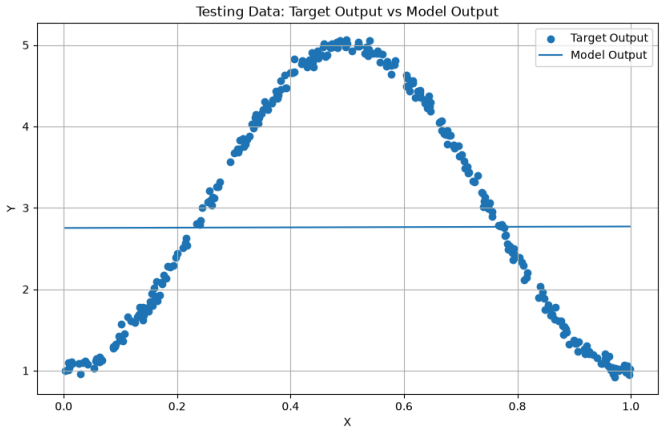


Fig. 13 - Testing Target vs Model Output Plot

- **Target Output vs Model Output Scatter Plots**
  - **Inference**
    - Both scatter plots show that the model output is concentrated in a very narrow range around 2.75-2.77.
    - The target values, however, vary approximately from 1 to 5 indicating that points do not follow a strong linear relationship like (y = x).
    - This confirms that the model predictions and actual target values have poor agreement.
    - The testing scatter plot shows behaviour similar to the training scatter plot, confirming consistent but limited model performance.



Fig. 14a - Training Target output vs Model output

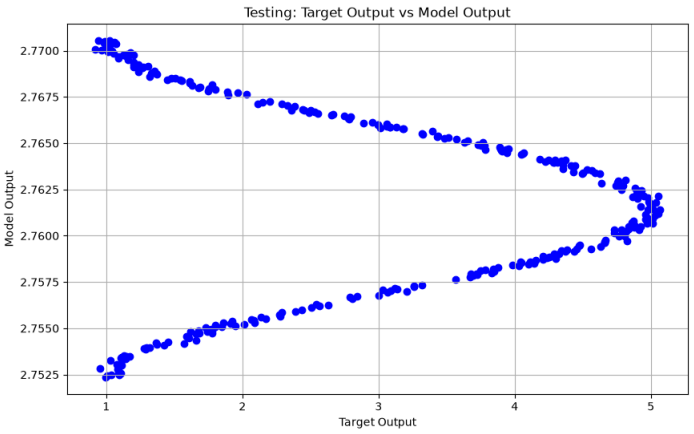


Fig. 14b - Training Target output vs Model output

- **Overall Inferences**
    - The model converges rapidly, but its predictions remain almost constant because the learned linear relationship cannot estimate the curved relationship present in the dataset.
    - The training RMSE (1.43098) and testing RMSE (1.43866) are close, indicating consistent performance on unseen data. However, the high %RMSE values of approximately 47.84% and 47.69% indicates that the model has substantial prediction errors.
    - Therefore, the main observation is that the perceptron implementation converges successfully, but a linear activation model is insufficient to accurately estimate the nonlinear trend present in the given dataset.

Dataset - 2 (Bivariate)

- **Final Model Weights**
    - The final weights obtained after training were:
      Final weights:
      `[2.80606148, -5.27873806, 1.06552957]`
    - Thus, the learned regression function is approximately:
      $\hat{Y}$ = 2.80606148 - 5.27873806*X_1 + 1.06552957*X_2
    - The model predicts the target using a **linear plane** in the three-dimensional (X_1-X_2-Y) space.

- **Training and Testing Results**

Dataset	RMSE	% RMSE
Training	1.3730	269.3519%
Testing	1.3611	269.3519%

Table 10 - Training and Testing results of bivariate dataset

    - The training and testing RMSE values are relatively close, indicating similar model behaviour on both datasets.

    - However, the very high %RMSE values indicate that the prediction error is large relative to the mean target value. Therefore, the linear perceptron does not accurately represent the target relationship as it tries to fit a linear hyperplane to non-linear data.

- **Average Training Error vs Epochs**
    - **Inference:**
        - The average error decreases sharply during the initial epochs.
        - After the initial few epochs, the average error becomes almost constant at approximately 0.93 and upon further epochs very less improvement is observed.
        - This indicates that the model converges very rapidly.

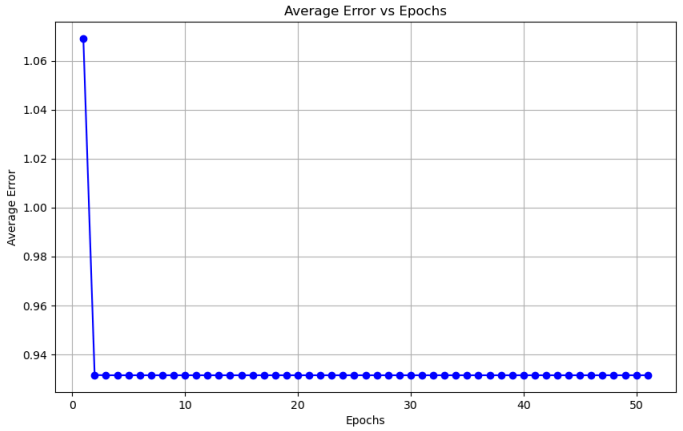


Fig. 15 - Average training error versus epochs

- **Training Regression Surface**
  - **Inference**:
    - The blue points represent the actual target values in the (X_1-X_2-Y) space.
    - The regression surface represents the output of the learned linear parameters of the perceptron.
    - The actual target data forms a non-planar pattern, while the model produces a flat plane surface.
    - A noticeable separation exists between many target points and the regression plane.
    - Hence, the linear model is unable to estimate the non-linear variation of output (Y) with respect to inputs (X_1) and (X_2).

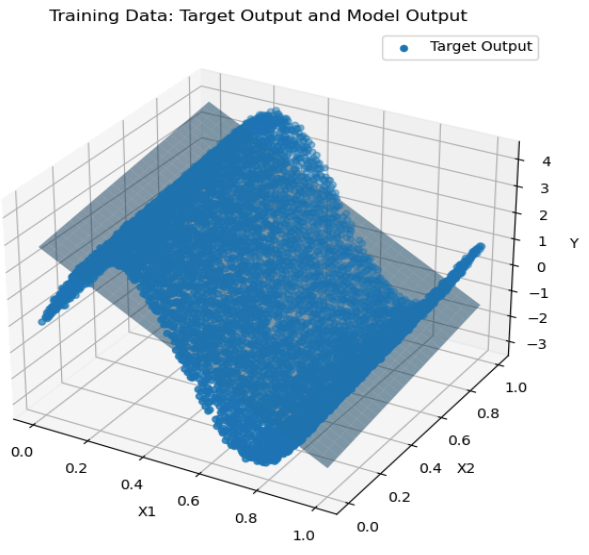


Fig. 16 - Training data: Regression Surface

- **Testing Regression Surface**
  - **Inference:**
    - The testing data has a similar trend as of the training data.
    - The learned model remains a linear plane across the (X_1-X_2) input space.
    - A considerable difference can be observed between the actual target points and the fitted plane. Therefore has limited predictive capability for the given bivariate relationship.
    - Similar behaviour in training and testing suggests that the high error is mainly due to the limitation of the linear model.

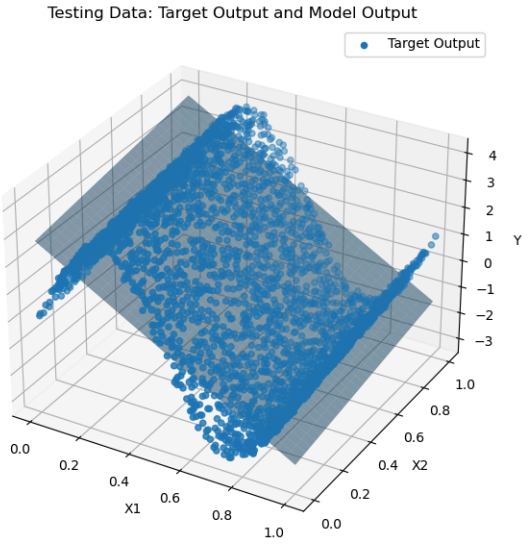


Fig. 17 - Training data: Regression Surface

- **Target Output vs Model Output Scatter Plots**
  - **Inference**
    - The target values range from over (-3) to (4).
    - The predicted values also show variation, but the points form broad bands rather than a close one-to-one relationship.
    - The model does not closely follow the expected linear relationship.
    - Significant prediction deviations are visible for both lower and higher target values.
    - The training and testing scatter plots have similar distributions, indicating consistency but poor predictive performance.

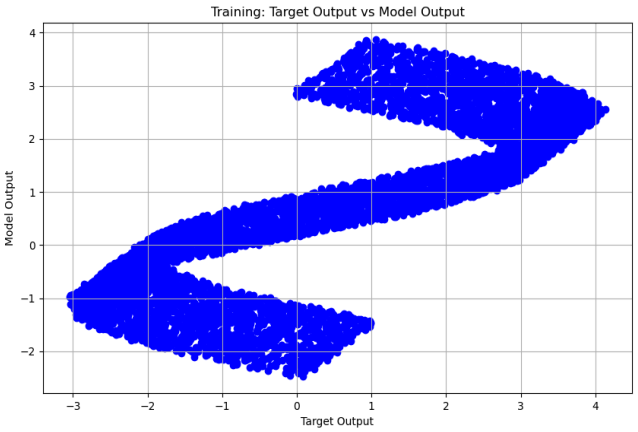


Fig. 18a - Training Target output vs Model output

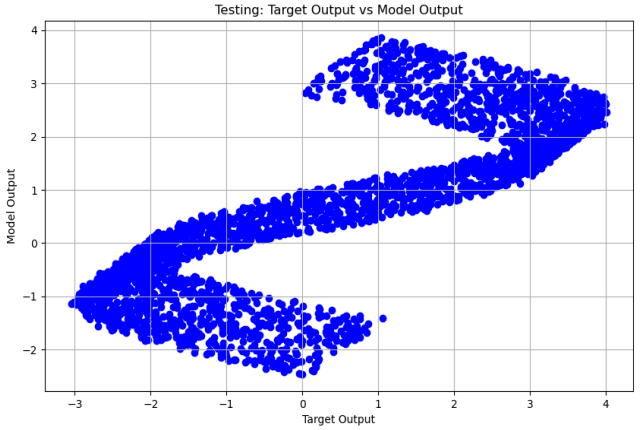


Fig. 18b - Training Target output vs Model output

- **Overall Inferences**
  - The bivariate perceptron regression model successfully learns a set of weights using gradient descent and converges very rapidly. However, the learned model represents the relationship using only a linear regression plane.
  - The training RMSE is 1.37304, while the testing RMSE is 1.36114 which indicates similar performance on training and unseen data. However, the %RMSE values of 269.35% and 285.28% are very high which indicates substantial prediction error relative to the mean target value.
  - The regression surface plots clearly display that the actual data has a more curved structure that cannot be represented adequately by a single plane. Therefore, the major limitation is the linear nature of the perceptron regression model.

4. Conclusion

- The experiments conducted in this assignment demonstrate the behaviour and limitations of a perceptron-based learning model across four different problems: linearly separable classification, non-linearly separable classification, univariate regression, and bivariate regression. The models were implemented from scratch using gradient descent, allowing the learning process and its dependence on the underlying data structure to be directly analysed.

- For the linearly separable classification problem, the One-vs-One strategy successfully extended the binary perceptron to the three-class problem. The decision-region plots showed clear linear boundaries between the classes, and all 450 test samples were classified correctly. Both logistic and tanh activation functions achieved 100% test accuracy, with precision, recall, and F-score equal to 1.00 for all three classes. The tanh model also produced lower final pairwise training errors in the reported experiment. These results confirm that a linear perceptron is highly effective when the classes can be separated using linear boundaries.

- For the non-linearly separable classification problem, the pairwise decision regions showed that the perceptron attempted to separate the curved class distributions using linear boundaries. This resulted in classification errors, particularly for Class 1. Logistic activation achieved 86.22% accuracy, whereas tanh achieved 88.00% accuracy, indicating a slight advantage for tanh. However, the difference between the two activation functions was relatively small compared with the fundamental limitation imposed by the linear decision boundaries. Thus, changing the activation function alone cannot overcome the inability of a single-layer perceptron to model complex non-linear class boundaries.

- For the univariate regression experiment, the model converged rapidly, but the learned output remained almost constant around 2.75–2.77 while the actual target followed a clear curved pattern. The training and testing RMSE values were 1.43098 and 1.43866, respectively, with %RMSE values of approximately 47.84% and 47.69%. The similar training and testing performance indicates that the poor result was not primarily caused by overfitting; instead, the linear perceptron lacked sufficient capacity to represent the non-linear relationship between the input and target.
  For the bivariate regression experiment, the perceptron learned a linear regression plane and converged rapidly. However, the actual target data exhibited a more complex, non-planar structure that could not be adequately represented by the fitted plane. The training and testing RMSE values were approximately 1.3730 and 1.3611, while the %RMSE values were very high, indicating substantial prediction error relative to the target scale. The regression-surface and scatter plots further confirmed the gap between the actual non-linear surface and the linear model output.
  Overall, the assignment demonstrates that successful convergence does not necessarily imply successful learning of the underlying problem. The perceptron converged rapidly in all the experiments, but its performance depended strongly on whether the target relationship could be represented by the model's linear structure. For classification, linear decision boundaries were sufficient for the linearly separable dataset but inadequate for the non-linearly separable dataset. Similarly, for regression, a linear perceptron could not accurately model the curved univariate relationship or the non-planar bivariate surface. Therefore, the major limitation observed in the experiments is model capacity rather than insufficient training time. More expressive models involving non-linear hidden layers or suitable feature transformations would be required to capture such non-linear relationships effectively.